

The Impact of Urbanization on Flooding

Overview

As areas become more urbanized, the risk of flooding increases. This is due to there being more impervious area and less vegetation, which leads to less infiltration, evapotranspiration, and interception, which all works to increase runoff (Mehltretter, 2026). The increase in runoff increases streamflow which decreases the amount of water storage and increases transport thus increasing the risk of flooding (Mehltretter, 2026). Flooding is a major problem in urbanized areas because it causes property damage and loss of life. The goal of this project is to create a model that can predict the flood risk based on urbanization level, precipitation, and river flowrate. The future objective of this model is for people to use it to predict the possibility of flooding, based on their location, the flowrate in the river, and the precipitation. The methodology used to create this model includes identifying a study area, identifying all necessary datasets, implementing all datasets into one file, splitting the data into three sets, training, testing, and validation, using random forest classifier to train the model, and using a confusion matrix to evaluate the model.

A supervised model was used to predict flood risk based on increased urbanization and precipitation. A supervised model was utilized because all the datasets were already known. With the model having all the data it created a function that can predict an output based on any new information added to the model. The output of the model is a determination of the level of flood risk, based on urbanization level, the river flow rate, and the precipitation. The important libraries used to create the model include sklearn and geopandas.

Methodology

The study area selected to create the model is the Rio Grande River in New Mexico. Data was collected from five cities and towns which are Ohkay Owingeh, Espanola, Corrales, Rio Rancho, and Albuquerque. The data collected for each town includes population, land use, flowrate, precipitation, and flood risk. The population and land use data were used to identify the level of urbanization and create the samples for the model. The flowrate data was used to identify the peak discharge in the Rio Grande river for the samples. The precipitation data was used to identify the total rainfall, which contributes to surface runoff, for the samples. The flood risk data was used as the target for the model and was used to determine the level of flood risk.

Population data was obtained using a shapefile from New Mexico Resource Geographic Information System, where each town and city is split into census tracts. Census tracts are small areas of towns and cities that are broken up based on census information, like population. Using this shapefile a total of 20 samples were obtained and used for the model. A map detailing where each of the samples are, is provided in Figure 1 below.



Figure 1: Map of census tracts used for model (Source: ArcGIS).

As seen in Figure 1 above, 2 samples were taken from the Ohkay Owingeh and Espanola region and the other 18 samples were taken from the Albuquerque, Corrales, and Rio Rancho region. This is due to there being a much higher population in the latter region. The total population of each sample ranges from 1968 people to 9017 people. This was the first set of data added to the model.

The next set of data added to the model was the land use data. This data gives further information into the level of urbanization for each area selected. The land use data was also obtained using a shapefile from New Mexico Resource Geographic Information System. The important information obtained from this shapefile was the “LUCODE.” The “LUCODE” defines how the land is being used. For example, a residential area has an “LUCODE” of 11. This data allowed the 20 samples that were taken to be split up even further based on the land use. To make this data usable for the dataset, the data needed to be transformed from categorical data to a numeric array, which was done using “pandas.get_dummies.” Once this was done, the land use data was joined with the population data.

The next dataset added was the flowrate data, which was obtained using the USGS National Water Dashboard. Due to the close proximity of Ohkay Owingeh and Espanola, the same station was used for the 2 samples relating to these towns. Also, due to the close proximity of Corrales, Rio Rancho, and Albuquerque, the same station was used to obtain the flowrate data for the other 18 samples. Flowrate data for the Ohkay Owingeh and Espanola region was collected from the years 2000 to 2025, and for the Corrales, Rio Rancho, and Albuquerque region, flowrate data was collected from 2004 to 2024. The maximum flowrate from each year was recorded and plotted, this can be seen in Figure 2a and Figure 2b below.

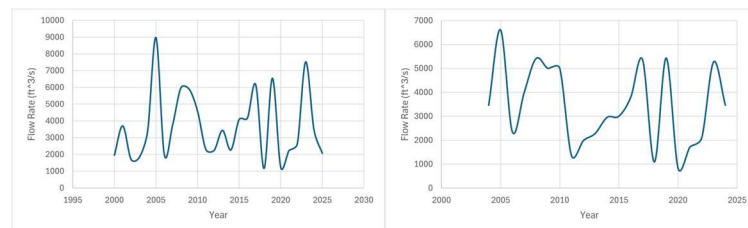


Figure 2a/2b: Graph of maximum flowrates for Ohkay Owingeh and Espanola (left), and Corrales, Rio Rancho, and Albuquerque (right) (Source: Excel).

Based on Figure 2a above, the highest value for the maximum flowrate is 8970 cfs. This value was used in the dataset for the 2 samples relating to the Ohkay Owingeh and Espanola region. Based on Figure 2b above, the highest value for the maximum flowrate is 6610 cfs. This value was used in the dataset for the 18 samples relating to the Corrales, Rio Rancho, and Albuquerque region. The maximum flowrate was used, because the maximum flowrate occurs during flooding events, so the maximum flowrate will be most helpful in determining flood risk.

The precipitation data was obtained from the National Oceanic and Atmospheric Administration. Once again, due to the samples' close proximity, the same precipitation data was used for samples in Ohkay Owingeh and Espanola, and the same precipitation data was used for samples in Corrales, Rio Rancho, and Albuquerque. For Ohkay Owingeh and Espanola, the maximum precipitation for each year was taken from 2006 to 2026, and for Corrales, Rio Rancho, and Albuquerque, the maximum precipitation for each year was taken from 2012 to

2026. The maximum precipitation from each year was recorded and plotted, which can be seen in Figure 3a and Figure 3b below.

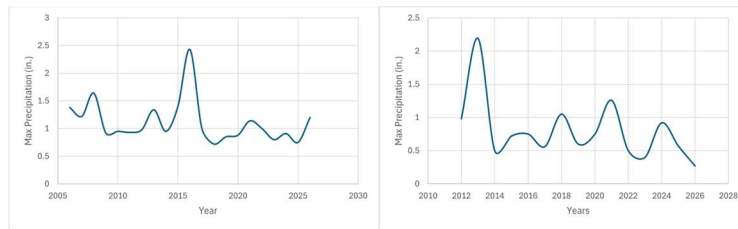


Figure 3a/3b: Graph of maximum precipitation for Ohkay Owingeh and Espanola (left), and Corrales, Rio Rancho, and Albuquerque (right) (Source: Excel).

From these 2 graphs, the highest value for maximum precipitation was taken from each and applied to the training dataset. Based on Figure 3a, the Ohkay Owingeh and Espanola samples have a maximum precipitation of 2.43 inches, and based on Figure 3b, the Corrales, Rio Rancho, and Albuquerque samples have a maximum precipitation of 2.19 inches. The maximum precipitation was used, because the highest amount of rainfall is most likely to occur during flooding events.

The final data added to the dataset was the flood risk data, which was obtained using FEMA’s National Flood Hazard Layer. This data layer defines the flood zones the samples are in. The dataset classifies flood zones using letter groups like, “AE,” “VE,” “A,” “AO,” “AH,” and “X.” To add this data as the target for the model, the flood zones were grouped together in risk groups. Areas with a flood zone categorized as “AE,” “VE,” “A,” “AO,” and “AH” were treated as the high flood risk group. Areas with a flood zone of “X” that also had a zone subtype of “0.2 PCT Annual Chance Flood Hazard,” which means there is a 0.2% annual chance of flooding in the area, was assigned to the moderate flood risk group. Otherwise, if an area had a flood zone of “X” and did not have the “0.2 PCT Annual Chance Flood Hazard” zone subtype, the area was assigned to the low flood risk group. Finally, any other flood zones not specified that appeared in the data were assigned to the unknown flood risk group. Once the risk groups were created, they were added to the training dataset as the target and were transformed from categorical variables using “Label Encoder” from sklearn. Once all the data was added, the model was ready to be trained and tested.

The predictors for the model include population, land use, river flowrate, precipitation, and flood zone, and the target was the risk group. 80% of the data was used for training the model, 10% was used for validation, and 10% was used for testing. The data used for training helps the model to learn the parameters, the data used for validation helps to tune hyperparameters, and the data used for testing helps to evaluate the performance of the model. The Random Forest Classifier from sklearn was used to train the model. Random Forest Classifier uses decision trees to make predictions. “Each tree looks at different random parts of the data, and their results are combined by voting for classification” (GeeksforGeeks, 2026). To evaluate the model a confusion matrix was used, which compared the true test values with the predicted test values. The confusion matrix when using 80% of the data for training is shown in Figure 4 below.

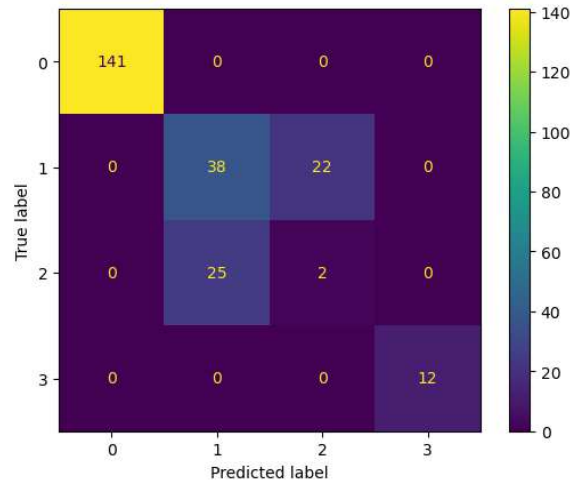


Figure 4: Confusion matrix when 80% of the data is used for training (Source: VSCode).

Results

In the confusion matrix above, 0 represents high flood risk, 1 represents low flood risk, 2 represents moderate flood risk, and 3 represents unknown flood risk. Based on the matrix above, the model correctly predicted 141 out of 141 samples relating to high flood risk and 12 out of 12 samples relating to unknown flood risk. However, when predicting low flood risk the model only correctly predicted 38 out of 63 samples and when predicting moderate flood risk only 2 out of the 24 samples were correctly predicted. This means the model correctly predicted all samples for 2 out of the 4 risk groups, high flood risk, and unknown flood risk. This also means that the model is less accurate when predicting low and moderate flood risk.

Conclusion

While the model was good at predicting high and unknown flood risk, the model faced challenges when predicting low and moderate flood risk. One limitation faced when creating the model was the limited amount of data from the study area. When obtaining the data for the Rio Grande River, some locations along the river had no usable data, which limited how many samples could be made for the model. To improve the model in the future, other study areas, like the Mississippi River, should be utilized, so more samples can be created. Also, the model was limited by the imbalance of samples, because most of the samples were for high flood risk and the number of samples for the other risk groups were much smaller. This means that even though the model correctly predicted all samples for unknown flood risk, due to the small number of unknown flood risk samples, the model may struggle in the future when trying to accurately predict unknown flood risk. To address this, more samples should be taken from more study areas, so the model is more balanced with samples from all different risk groups. Finally, the model can further be improved by adding an input that determines the distance a sample is from a river. This would help the model more accurately predict flood risk, based on the distance to the river. With these improvements to the model, the model will more accurately predict flood risk and help people to determine if they are located in a high flood risk area.

References

- ArcGIS Web Application*. (n.d.). Hazards-Fema.maps.arcgis.com. <https://hazards-fema.maps.arcgis.com/apps/webappviewer/index.html?id=8b0adb51996444d4879338b5529aa9cd>
- Browse rgis data*. (n.d.). Rgis.unm.edu. <https://rgis.unm.edu/rgis6/>
- GeeksforGeeks. (2024, February 22). *Random Forest Algorithm in Machine Learning*. GeeksforGeeks. <https://www.geeksforgeeks.org/machine-learning/random-forest-algorithm-in-machine-learning/>
- Google Earth. (2025). *Google Earth*. Google.com; Google. <https://earth.google.com/web>
- Map Viewer*. (2026). Arcgis.com. <https://sunny-buffalo.maps.arcgis.com/apps/mapviewer/index.html?webmap=ef14548f0aea4a70908d02229a9b149b>
- Mehltretter, S. (2026, February 3). *Fluvial Systems & Hydrology* [Review of *Fluvial Systems & Hydrology*].
- OpenAI. (2025). *ChatGPT*. ChatGPT; OpenAI. <https://chatgpt.com/> (ChatGPT was only used to help create parts of the model).
- Past Weather | National Centers for Environmental Information (NCEI)*. (n.d.). Wwww.ncei.noaa.gov. <https://www.ncei.noaa.gov/access/past-weather/>
- United States Census Bureau. (2023). *Explore Census Data*. United States Census Bureau. <https://data.census.gov/>
- USGS | National Water Dashboard*. (n.d.). Dashboard.waterdata.usgs.gov. <https://dashboard.waterdata.usgs.gov/app/nwd/en/>